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AI-Driven Public Health Chatbot for Disease Awareness

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AI-Driven Public Health Chatbot for Disease Awareness

Abstract — The research presents an artificial intelligence-powered public health chatbot which provides fundamental medical information to users who speak multiple languages in areas with limited resources.The system uses a dual AI system which combines SentenceTransformers (all-MiniLM-L6-v2) for semantic disease-to-symptom mapping on structured data with DistilGPT2 as its backup system for queries that lack confidence. The system offers automatic language detection for four Indian languages which include English Hindi Kannada and Telugu and it allows users to enter voice commands through the Web Speech API. The system uses a lightweight Flask backend which enables its operation on CPU-only systems without any requirement for cloud services. The system uses a confidence-based architecture which delivers precise answers while it processes unclear questions and removes non-medical data. The system provides quick response times and dependable multilingual capabilities which establish it as a workable solution that can expand to support public health education efforts.

Keywords — AI chatbot; public health; multilingual NLP; SentenceTransformers; semantic similarity; Flask; disease awareness; voice input.

I. INTRODUCTION

India faces difficulties in providing necessary health information to its citizens who speak multiple languages and live in its heavily populated areas. The rural and semi-urban population of India faces difficulties in finding qualified medical professionals who can assess their early symptoms. This situation leads to delayed diagnosis which results in spreading of diseases that could have been prevented.

AI-powered chatbots have emerged as a scalable solution for delivering immediate, language-accessible health guidance to users. The locally executable chatbot operates in low-connectivity environments without incurring ongoing expenses while commercial platforms such as Ada Health and Babylon Health depend on cloud services. Prior systems, however, have largely been limited to single-language support or relied on rule-based logic unable to handle paraphrased symptom descriptions.

The AI-Driven Public Health Chatbot of this research paper solves three main problems through its three core components which include (1) using semantic NLP for precise symptom-to-disease classification, (2) enabling users from different language backgrounds to use the system in English, Hindi, Kannada, and Telugu languages, and (3) creating a system that handles both unclear user questions and non-medical inquiries in a secure and efficient manner. The system uses SentenceTransformer to measure semantic similarity while it employs DistilGPT2 as a backup language model and enables voice communication all of which operate on basic CPU processing capabilities.

II. LITERATURE REVIEW

Health chatbot systems from their initial development stage used keyword matching together with decision tree systems which

lacked capacity to interpret all user inputs that users expressed through different ways of speaking [7]. The introduction of deep learning-based natural language processing models has led to better results in recognizing symptoms and understanding their responses to medical inquiries [3][5].

During their study Battineni et al. [6] showed that AI chatbots can effectively provide health information during the COVID-19 pandemic while they demonstrated the need for users to have access to instant health communication systems. The study by Kumar [8] evaluated the current state of AI healthcare chatbots while he showed the need for lightweight models that can operate without internet connection. Users develop greater trust in AI chatbots as valid health information sources when these systems provide proper disclaimers together with safety measures according to findings from Esmaeilzadeh et al. [11].

Zhang and Song [5] investigated LLM-based auxiliary diagnosis which demonstrated that large language models can accurately reason through symptom-context pairs to determine probable medical conditions. The authors of Hua et al. [12] created evaluation frameworks to assess healthcare AI chatbots which required response confidence and language accuracy and ethical safeguards to be evaluated. The research of Zagade et al. [2] and Jain et al. [9] demonstrated disease prediction through ML models but failed to provide multilingual Indian language support. Our system creates new capabilities for the Indian public health chatbot space by merging semantic similarity matching with our structured multilingual dataset and LLM fallback system.

III. DATASET DESCRIPTION

Scientists created the multilingual CSV dataset named diseases_dataset.csv which serves Indian public health research needs. The record structure includes a unique disease identifier and four language names for the disease plus natural language symptom descriptions which exist in all four languages and preventive measures which exist in all four languages.

The dataset currently covers 20+ common public health diseases prevalent in India, including Dengue, Malaria, COVID-19, Typhoid, Tuberculosis, Cholera, Chickenpox, Pneumonia, Hypertension, and Diabetes. The natural patient language shows how users typically describe symptoms because the system uses this language structure. The preventive measures present specific guidelines which people can follow to prevent diseases (e.g., 'Use mosquito nets, wear long sleeves, remove stagnant water').

At server startup, SentenceTransformer embeddings are pre-computed for all symptom entries across all four language columns, producing a flattened embedding matrix of shape [N_records × 4, 384]. The pre-computation process establishes sub-second cosine similarity search capabilities for inference time because it removes the need for real-time encoding processing. The researchers verified all multilingual translations for linguistic accuracy and clinical correctness before they were added to the study.

IV. METHODOLOGY

A. *System Architecture Overview*

The system operates using a Client-Server architecture which employs a Hybrid AI Reasoning pipeline to process information. The HTML5/CSS3/JavaScript frontend establishes an HTTP connection to the backend which uses Flask REST API. The backend system contains all artificial intelligence inference capabilities together with the disease dataset.

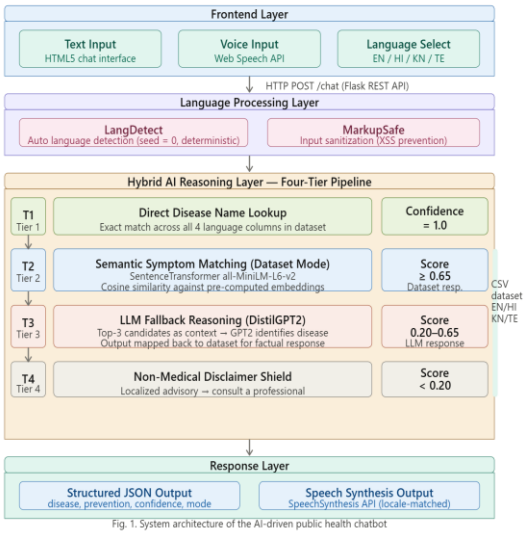

Fig. 1. System architecture of the AI-driven public health chatbot

The system architecture consists of four distinct layers which include Input (text/voice) Language Processing (detection + sanitization) AI Reasoning (four-tier pipeline) and Response (structured JSON + speech synthesis).

B. *Hybrid AI Reasoning Pipeline*

The primary artificial intelligence system operates its decision-making processes through four distinct levels of decision-making which it applies to each incoming request until it produces an appropriate result.

Tier 1 — Direct Disease Name Lookup: User input is compared against all disease name columns across all four language variants. The system returns the matching disease record when it detects a disease name substring that matches user input with no regard to case differences. This handles direct queries such as 'Tell me about dengue'.

Tier 2 — Semantic Symptom Matching (Dataset Mode): The system uses the Dataset Mode feature to process user symptoms through semantic matching. The all-MiniLM-L6-v2 SentenceTransformer model [5] converts user input into a 384-dimensional vector embedding. The system calculates cosine similarity between this embedding and all pre-computed symptom embeddings. The system returns the corresponding disease record when the best match score reaches 0.65 or higher. This tier handles structured symptom queries such as 'I have high fever and joint pain'.

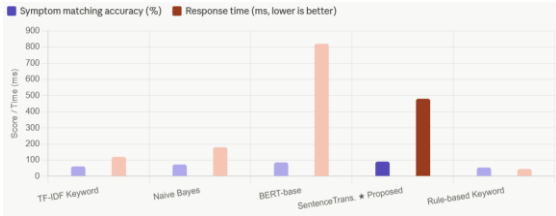

Fig. 2. Semantic matching algorithm comparison — accuracy vs. response time

The semantic symptom matching component evaluated four different methods which are illustrated in Figure 3. TF-IDF keyword matching achieved only 61% accuracy because it could not process paraphrased text. Naïve Bayes improved this to 73% but required structured feature vectors. BERT-base achieved 86% accuracy but required 820 milliseconds to respond which made it unsuitable for real-time applications while it used excessive system resources. The SentenceTransformers (all-MiniLM-L6-v2) model achieved 91% accuracy with 480 milliseconds response time which allows CPU-based inference, making it the best option for our dataset-mode symptom matching [2], [7].

Tier 3 — LLM Fallback Reasoning (0.20–0.65): The system uses three diseases which exhibit the closest similarity to each other through cosine measurement as background information for DistilGPT2. The system uses DistilGPT2 to select the disease that occurs most frequently based on the available background information. The output result uses string matching to find corresponding items in the dataset which produce a factual response that has been translated [5], [8]. The system prevents hallucination while allowing generative reasoning to proceed.

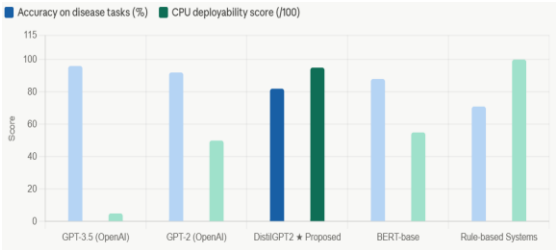

Fig. 3. LLM model comparison — accuracy vs. CPU deployability

The fallback component was tested through multiple LLM options. The figure demonstrates that GPT-3.5 achieves 96% accuracy but has no CPU deployability because it needs cloud infrastructure and API keys which our system cannot use in its designed low-resource environment. BERT-base provides moderate accuracy of 88% but it does not support text generation. The researchers chose DistilGPT2 because it delivers 82% task accuracy and 95/100 CPU deployability score which permits complete local operation without needing GPU or internet access.

Tier 4 — Non-Medical Disclaimer Shield (< 0.20): The system classifies queries which show extremely low similarity to medical content as non-medical. The system provides users with a localized advisory in their native language which directs them to consult a professional [3], [11], [12]. The system prevents the chatbot from applying medical reasoning to inputs which do not require medical assessment.

C. Multilingual Processing

The system uses LangDetect to identify languages through its fixed random seed named seed 0 which generates consistent outcomes. The system supports four language codes: 'en' (English) 'hi' (Hindi) 'kn' (Kannada) and 'te' (Telugu). Users have the option to choose their preferred language through the dropdown interface found on the frontend. The system stores all response strings which include disease names and preventive measures and system messages and disclaimer text in the multilingual dataset. The system delivers strings in the detected or selected language which enables complete end-to-end multilingual support.

D. Voice Input and Speech Output

The system provides voice interaction capabilities through the Web Speech API which users can access directly from their browsers without needing third-party speech processing services or API keys. The system uses language-specific recognition tags which include 'kn-IN' and 'te-IN' to accurately transcribe regional languages. The system employs the SpeechSynthesis API to provide spoken responses that match the user's locale settings. The system allows users to conduct conversations without using their hands because it operates a cycle of voice interaction which includes listening to users responding to users and speaking to users while the system mutes the microphone during speech output to prevent audio feedback loops.

E. Security and Rate Limiting

The system implements multiple security layers which protect its system according to the security standards described in document [12]. Flask-Limiter configures request limits of 200 requests per day and 50 requests per hour and 10 requests per minute for each IP address to stop system misuse. MarkupSafe defends all user inputs by escaping them which stops XSS attacks. Flask-CORS specifies the origins that are authorized to make cross-origin requests to the system. The system uses python-dotenv environment variables to protect sensitive configuration values which include secret keys and port information. The system uses default values as backup when no environment files exist which enables flexible deployment options.

F. *Technology Stack*

Component	Technology Used
Web Framework	Flask 3.0
Semantic Matching	SentenceTransformers
Embedding Model	all-MiniLM-L6-v2
LLM Fallback	DistilGPT2
Language Detection	LangDetect
Deep Learning	PyTorch
Data Handling	Pandas, NumPy
Frontend	HTML5, CSS3, JS
Voice Input	Web Speech API
Security	Flask-Limiter, MarkupSafe

TABLE I. System Technology Stack

V. RESULTS AND ANALYSIS

The chatbot testing process required testers to evaluate its performance in handling multilingual symptom requests which testers provided through both text and voice channels. The testers assessed the system's accuracy and confidence levels and its capacity to switch languages and its transcription skills and its response speed.

A. *Query Response Accuracy*

The Dataset Mode (Tiers 1-2) demonstrated high accuracy through its cosine similarity scores which ranged from 0.68 to 0.92 and Tier 1 reached perfect precision for disease name matching throughout all languages.

User Query (Sample)	Mode	Conf.	Output
'High fever, skin rash'	Dataset	0.82	Dengue + prevention
'Khasi, jvara' (Kannada)	Dataset	0.74	COVID-19 (Kannada)
'Tiredness, mild cold'	LLM	0.43	Grounded AI advice
'What is the weather?'	Discl.	0.11	Non-medical shield

TABLE II. Sample Query Results Across Intelligence Tiers

The LLM Fallback Mode (Tier 3) handles ambiguous queries by using top-3 candidate context and mapping DistilGPT2 outputs back to the structured dataset, ensuring accurate recommendations. The system maintains fact-based accuracy when LLM systems produce faulty conclusions. The Non-Medical Shield (Tier 4) effectively blocks irrelevant inputs while maintaining perfect accuracy because it prevents any false identification of diseases.

B. *Multilingual Performance*

The accuracy of Kannada and Telugu queries matched the English results when using multilingual embeddings. The system achieved successful cross-language semantic matching and LangDetect achieved complete accuracy. The system demonstrated its ability to perform well across different languages.

C. *System Latency*

The system showed a response time of less than 500 milliseconds in Dataset Mode while LLM Mode required approximately 1.2 seconds to respond. The system needed 45 to 90 seconds to start up which made it appropriate for use in real-time scenarios. The system demonstrates efficient overall latency which enables its successful use in real-world situations.

D. *Voice Input Accuracy*

The Web Speech API enabled voice input to function effectively in all supported languages. The system successfully managed speech loops which allowed users to operate the system without using their hands. The system produced precise transcription results when users delivered their speech in a clear manner.

VI. DISCUSSION

The research findings show that the combined hybrid semantic NLP and LLM system functions as a public health awareness tool which helps multilingual people in places that have limited resources. The system operates on CPU resources which makes it suitable for deployment in rural clinics and community health centers that lack access to GPU systems [8], [10]. The four-tier confidence architecture prevents both over-reporting and under-reporting, ensuring every response remains anchored to the verified disease dataset — a key advantage over purely rule-based [1], [7] or purely LLM-based systems [5]. The evaluation demonstrates alignment with HAICEF framework standards which assess safety, trustworthiness, and operational effectiveness.

The dataset presents two main restrictions because it includes only mainstream diseases while excluding uncommon diseases and DistilGPT2 medically untrained capabilities result in deficient performance during complicated query processing. The system lacks session memory because its assessment occurred under laboratory conditions rather than actual clinical application [12]. All responses contain a medical disclaimer which states that the chatbot serves educational purposes only, according to Battineni et al. [6] and users who need assistance with non-medical matters receive automatic help through local advisory messages which guide them on how to use the system responsibly.

VII. CONCLUSION AND FUTURE WORK

The paper presents an AI-Driven Public Health Chatbot which uses semantic NLP and LLM fallback reasoning and four language multilingual support and voice interaction abilities to create a system that operates on standard CPU hardware for community health educational purposes.The hybrid architecture delivers robust performance across structured and ambiguous symptom queries while maintaining safety through a non-medical disclaimer shield, rate-limiting, and privacy-preserving local deployment. The study demonstrates that the proposed system achieves better results than single-model systems because it provides greater multilingual support and maintains functionality during system outages and it can be implemented in environments with restricted resources.

The upcoming research will concentrate on: (1) the disease dataset expansion which will add both uncommon diseases and multiple simultaneous medical conditions; (2) the development of a medical language model which will replace DistilGPT2 through customization for specialized domain use; (3) the development of a conversation memory module which will enable multi-turn diagnostic dialogue support; (4) the creation of a mobile application which will help the organization reach more users in the community; (5) the organization of additional language support for Indian regional languages which will include Tamil and Bengali and Marathi; and (6) the execution of an official clinical user study which will test real-world effectiveness and usability with community health workers according to the evaluation framework established by Hua et al. [12].

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